

Problem Statement & Solution Explainer

RetailFlow / BarrelFlow POS for retail and liquor-store operations

Purpose: This brief explains the business problem and how the proposed POS product solves it for small retail stores that need faster billing, cleaner inventory control, and clearer operational reporting.

Problem Statement

Small retail and liquor stores often run daily operations across disconnected tools: manual billing, spreadsheet inventory, informal customer tracking, and paper receipts. This creates slow checkout lines, inaccurate stock counts, limited sales visibility, and avoidable mistakes during busy store hours.

The core problem is operational fragmentation:

- Billing is slow when cashiers must manually calculate carts, discounts, taxes, and payment status.
- Inventory is unreliable when product sales do not immediately update stock and packaging quantities.
- Customer and order history are hard to retrieve when receipts, phone numbers, and previous purchases are spread across multiple places.
- Store owners lack real-time visibility into revenue, best-selling items, payment mix, and low-stock risks.
- Authentication and staff access need to be simple enough for store teams but controlled enough for business use.

Solution Overview

RetailFlow / BarrelFlow is a modern web-based POS workspace that brings billing, inventory, orders, customers, promotions, settings, and reporting into one interface. The product is built as a Next.js application with a polished cashier dashboard and Google authentication support.

Capability	How the solution responds
Fast checkout	POS cart, payment method selection, ID verification prompt, and receipt flow
Inventory control	Products, stock quantities, packaging units, categories, and printable tags
Customer history	Customer records built from sales, contact details, and order counts
Operational visibility	Dashboard, reports, revenue totals, payment mix, and recent orders
Secure access	Google login with Firebase plus email/password flows and reset support

How the Product Works

- Cashiers sign in, open the POS terminal, select or scan products, verify age when required, choose payment mode, and confirm the sale.
- Inventory screens let the store maintain products, categories, stock quantities, pricing, packaging units, and printable product tags.

- Orders, customers, and reports give owners a structured view of transactions, customer activity, sales totals, and operational trends.
- Settings allow the store to manage branding, receipt content, theme, employee details, and business preferences.

Small Diagram: Solution Flow

Store Staff	RetailFlow POS	Inventory + Orders	Customers + Receipts	Reports + Decisions
Logs in and selects products	Builds bill and captures payment	Updates stock and transaction history	Stores buyer context and prints receipts	Shows sales, stock, and next actions

Flow: Store Staff -> POS -> Inventory/Orders -> Customers/Receipts -> Reports/Decisions

Current State and Next Step

The current product is a strong working POS prototype with authentication and a complete front-end operating flow. Its business data layer currently relies heavily on a browser-local mock API for products, orders, customers, and reports. The recommended next step is to connect those modules to real Supabase database tables with row-level security so each store account has durable, isolated production data.

Outcome: The solution turns fragmented store operations into one faster, clearer workflow: sell products, update inventory, track customers, print receipts, and review performance from the same POS system.